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Department of Mathematics
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Math 101 Final Exam Term 251

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Let x and y be two positive numbers such that the sum of the first number cubed and the second number is 500, and the product of x and y is maximum. Then $x + y =$

(a) 385

(b) 240

(c) 120

(d) 380

(e) 375

2. Which of the following is/are True? (I) There is no function with an infinite number of critical numbers. (II) Every polynomial has at least one critical number. (III) A polynomial of degree $n \geq 1$ has at most $(n-1)$ critical numbers.

(a) I and III

(b) III

(c) I

(d) II and III

(e) II

3. Consider the function $f(x) = \sqrt[3]{x}$. Which statement is TRUE about $f(x)$?

(a) $f(x)$ has two critical points

(b) $f(x)$ has no inflection point

(c) $f(x)$ is decreasing for all $x > 0$

(d) $f(x)$ concave down on $(-\infty, 0)$

(e) $f(x)$ has one inflection point at $(0, 0)$

4. If $y = e^{-x/2}\cos 4x$, then $dy =$

(a) $e^{-x/2}\left(4\cos 4x + \frac{1}{2}\sin 4x\right)dx$

(b) $e^{-x/2}\left(4\cos 4x - \frac{1}{2}\sin 4x\right)dx$

(c) $e^{-x/2}\left(-4\cos 4x - \frac{1}{2}\sin 4x\right)dx$

(d) $e^{-x/2}\left(4\sin 4x - \frac{1}{2}\cos 4x\right)dx$

(e) $e^{-x/2}\left(-4\sin 4x - \frac{1}{2}\cos 4x\right)dx$

5. $\int \frac{x+6}{\sqrt{x}}\,dx =$

(a) $\frac{3}{2}x^{3/2} + 6x^{1/2} + c$

(b) $\frac{3}{2}x^{3/2} + 12x^{1/2} + c$

(c) $\frac{2}{3}x^{3/2} + 12x^{1/2} + c$

(d) $\frac{2}{3}x^{2/3} + 12x^{1/2} + c$

(e) $\frac{3}{2}x^{2/3} + 12x^{1/2} + c$

6. $\lim_{x\to 0^+}(\sin x)^x =$

(a) $\frac{\pi}{4}$

(b) DNE

(c) 1

(d) 0

(e) $\frac{e}{4}$

7. The graph of the function $f(x) = \dfrac{\cos x}{1+\sin x}$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

(a) is increasing and concave downward

(b) intersects the x-axis at $x = 0$

(c) is decreasing and concave upward

(d) is increasing and concave upward

(e) is decreasing and concave downward

8. $\lim_{x\to-\infty}\left(3x+\sqrt{9x^2-x}\right)=$

(a) $\frac{1}{6}$

(b) $\frac{1}{3}$

(c) $-\infty$

(d) ∞

(e) $\frac{1}{2}$

9. Consider the function $f(x)=\frac{x}{x^2+1}$. Which statement is TRUE about the graph of f?

(a) $f(x)$ has exactly one point of inflection at $(0,0)$

(b) $f(x)$ is increasing on $(-\infty,-1)$ and $(1,\infty)$

(c) $f(x)$ has a relative minimum at $\left(-1,-\frac{1}{2}\right)$ and a relative maximum at $\left(1,\frac{1}{2}\right)$

(d) $f(x)$ is concave upward on $(0,\sqrt{3})$ and concave downward on $(-\sqrt{3},0)$

(e) The line $x=0$ is a vertical asymptote of $f(x)$

10. $\lim_{x\to0^+}(e^x+x)^{2/x}=$

(a) ∞

(b) e^4

(c) e

(d) 1

(e) e^2

11. If f is a function such that $f''(x)=-2$, $f(-1)=3$ and $f(0)=1$, then $f(1)=$

(a) 1

(b) 0

(c) -2

(d) -1

(e) -3

12. $\lim_{x\to 0} \frac{\sinh 2x}{x} =$

(a) 0

(b) 1

(c) 2

(d) -2

(e) -1

13. The function $f(x) = \dfrac{\ln(x^2+1)}{x+1}$

(a) has two vertical asymptotes at $x = 1$ and $x = -1$

(b) has one vertical asymptote at $x = -1$

(c) has no vertical asymptote

(d) has one vertical asymptote at $x = 1$

(e) has two vertical asymptote at $x = 0$ and $x = -1$

14. The sum of all critical numbers of the function $f(x) = \dfrac{4x}{x^2+1}$ is equal to

(a) 2

(b) 0

(c) -1

(d) -3

(e) 1

15. If $f(x) = xe^{2x}$, then $f^{(4)}(1) =$

(a) $28e^2$

(b) $36e^2$

(c) $24e^2$

(d) $20e^2$

(e) $48e^2$

16. The sum of all the relative maxima and minima of the function $f(x) = \sin x \sinh x - \cos x \cosh x$ on $(-4, 4)$ is

(a) $2\cosh \pi - 1$

(b) $1 - \cosh \pi$

(c) $1 - 2\cosh \pi$

(d) $\cosh \pi - 1$

(e) -1

17. $\int (\sec y)(\tan y - \sec y)\, dy =$

(a) $\cos y + \tan y + c$

(b) $\sec y - \tan y + c$

(c) $\sec^2 y - \tan y + c$

(d) $\tan^2 y + \csc y + c$

(e) $\sec^2 y + \tan y + c$

18. If

$$f(x) = \begin{cases} x^2 & x \leq 2 \\ 8 - 2x & 2 < x < 4 \\ 4 & x \geq 4 \end{cases}$$

then

(a) $\lim_{x \to \infty} f(x) = \infty$

(b) f is increasing on $(2, 4)$

(c) $f'(2)$ does not exist

(d) $\lim_{x \to -\infty} f(x)$ exists

(e) f is continuous on $\mathbb{R}$

19. The function $f(x) = 7x^3 - 21x + 3$ is

(a) increasing on $(0, 2)$

(b) increasing on $(0, \infty)$

(c) decreasing on $(-1, 1)$

(d) increasing on $(-1, 1)$

(e) decreasing on $(-\infty, -1)$

20. An equation of the tangent line to the curve $x^{2/3} + y^{2/3} = 5$ at the point $(8, 1)$ is

(a) $y - x + 7 = 0$

(b) $y - 2x + 15 = 0$

(c) $2y - x + 6 = 0$

(d) $y + 2x - 17 = 0$

(e) $2y + x - 10 = 0$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$P = 500x - x^4$, $x = 5$, $y = 375$, so $x + y = 380$.
2	**(b)**	Only III is true: f' has degree $n-1$; $\sin x$ and x are counterexamples to I and II.
3	**(e)**	$f'' = -\frac{2}{9}x^{-5/3}$ changes sign at 0: inflection point $(0, 0)$.
4	**(e)**	$dy = y'\,dx$ with $y' = e^{-x/2}(-4\sin 4x - \frac{1}{2}\cos 4x)$.
5	**(c)**	$\int (x^{1/2} + 6x^{-1/2})dx = \frac{2}{3}x^{3/2} + 12x^{1/2} + c$.
6	**(c)**	$\ln L = x \ln \sin x \to 0$, so the limit is $e^0 = 1$.
7	**(c)**	$f' = -\frac{1}{1+\sin x} < 0$ and $f'' = \frac{\cos x}{(1+\sin x)^2} > 0$: decreasing, concave up.
8	**(a)**	Conjugate gives $\frac{-x}{\sqrt{9x^2-x}-3x} \to \frac{-1}{-6} = \frac{1}{6}$.
9	**(c)**	$f' = \frac{1-x^2}{(x^2+1)^2}$: min at $(-1, -\frac{1}{2})$, max at $(1, \frac{1}{2})$.
10	**(b)**	$\ln L = \lim \frac{2\ln(e^x+x)}{x} = 4$, so $L = e^4$.
11	**(e)**	$f = -x^2 - 3x + 1$, so $f(1) = -3$.
12	**(c)**	L'Hopital: $2\cosh 2x \to 2$.
13	**(b)**	Denominator zero only at $x = -1$, numerator $\ln 2 \neq 0$: one vertical asymptote $x = -1$.
14	**(b)**	$f' = \frac{4(1-x^2)}{(x^2+1)^2} = 0$ gives $x = \pm 1$, sum 0.
15	**(e)**	$f^{(4)} = (32 + 16x)e^{2x}$, so $f^{(4)}(1) = 48e^2$.
16	**(a)**	$f' = 2\sin x \cosh x$: min $f(0) = -1$, maxima $f(\pm\pi) = \cosh \pi$; sum $2\cosh \pi - 1$.
17	**(b)**	$\int (\sec y \tan y - \sec^2 y)dy = \sec y - \tan y + c$.
18	**(c)**	Left slope 4, right slope -2 at $x = 2$, so $f'(2)$ does not exist.
19	**(c)**	$f' = 21(x-1)(x+1) < 0$ on $(-1, 1)$, so f is decreasing there.
20	**(e)**	$y' = -(y/x)^{1/3} = -\frac{1}{2}$ at $(8, 1)$: $2y + x - 10 = 0$.